Nugget Line — process mapping report

C.H. Guenther & Son — Duncanville, TX · 11/13/2026 · Current state

This report contains DEMONSTRATION DATA. Values are illustrative and are not plant measurements. Values shown as NYC are Not Yet Collected. Each result is labeled Verified inputs, Estimate or Incomplete according to the status of its inputs. Estimated savings are not validated until Finance confirms them against actual cost results.

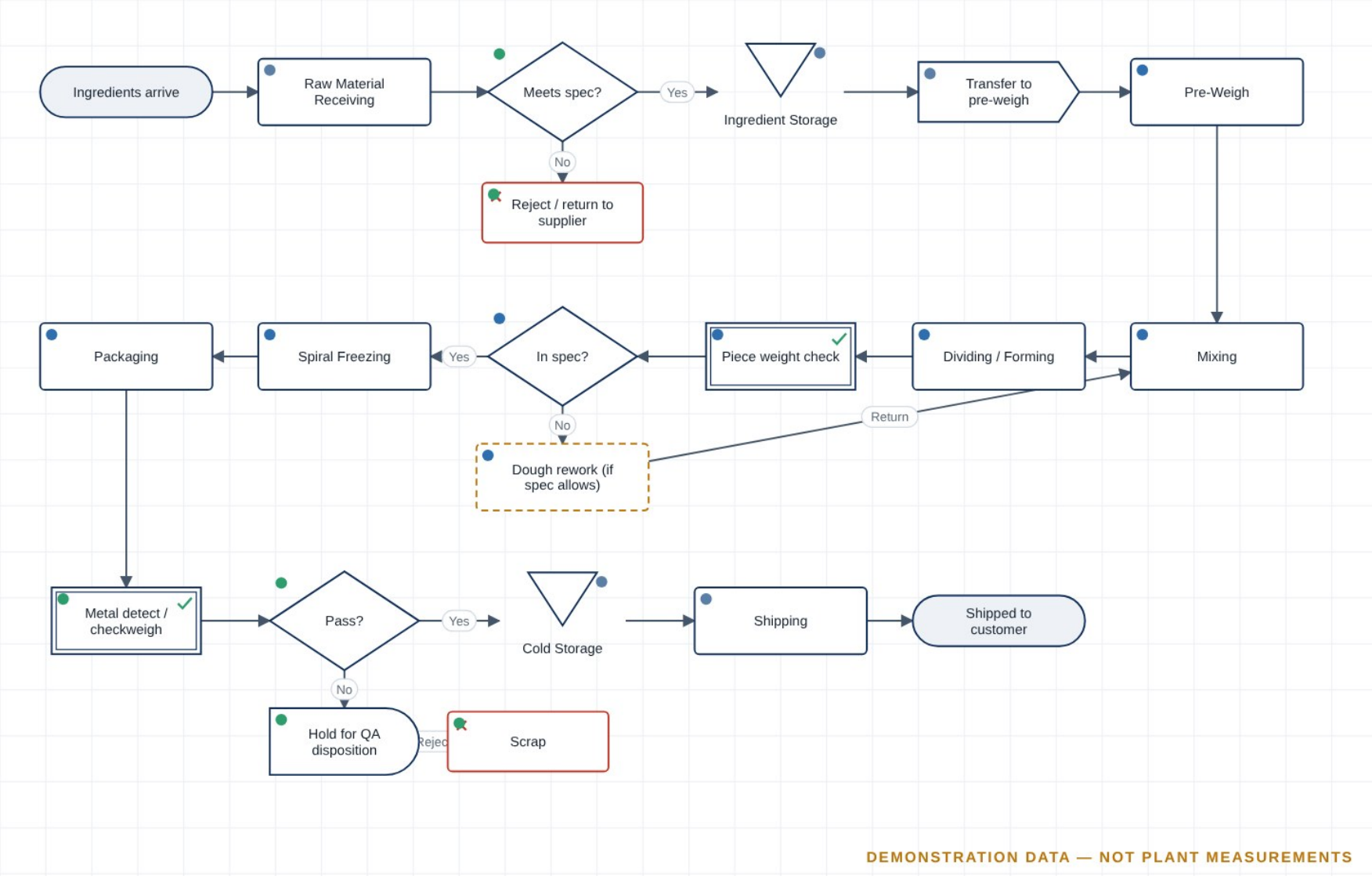
Key performance indicators

Calculated values with data status and missing inputs

KPI	Value	Data status	Missing inputs	Formula
Takt time	10.15 s	Estimate		Available time per day × 60 ÷ Customer demand per day
Total lead time	117.35 hr	Estimate		£ Queue time + £ Processing time over applicable step
Value-added time	49 min	Estimate		£ Processing time of steps classified Value-added
Process cycle efficiency	0.70%	Estimate		Value-added time ÷ Total lead time
Bottleneck effective C/T	7.87 s (Packaging)	Estimate		Cycle time ÷ Uptime. Where uptime is not collected, cycle time is used and the result is marked Estimate.
Line capacity	457 case/hr	Estimate		3600 ÷ Bottleneck effective cycle time
Rolled throughput yield	95.5%	Incomplete	Raw Material Receiving; Ingredient Storage; Cold Storage; Shipping	(Yield) over steps; where yield is missing, (1" Scrap
Changeover time	17 hr/week	Estimate		£ (Changeover time × Changeovers per week) ÷ 60
Travel labor	661 hr/yr	Estimate		Travel time per shift × People × Shifts/day × Operating days/year ÷ 60
Travel labor cost	$18,499	Estimate		Annual travel labor hours × Loaded labor rate

Process flow map

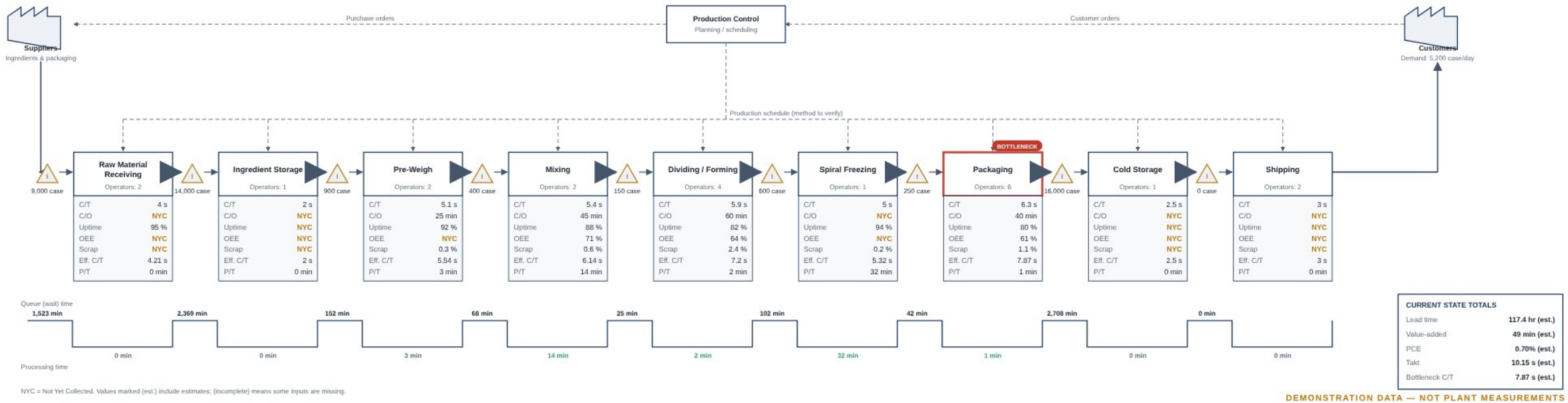
Operations, inspections, decisions, rework and scrap.



DEMONSTRATION DATA — NOT PLANT MEASUREMENTS

Value stream map

Material and information flow with data boxes and lead-time ladder.

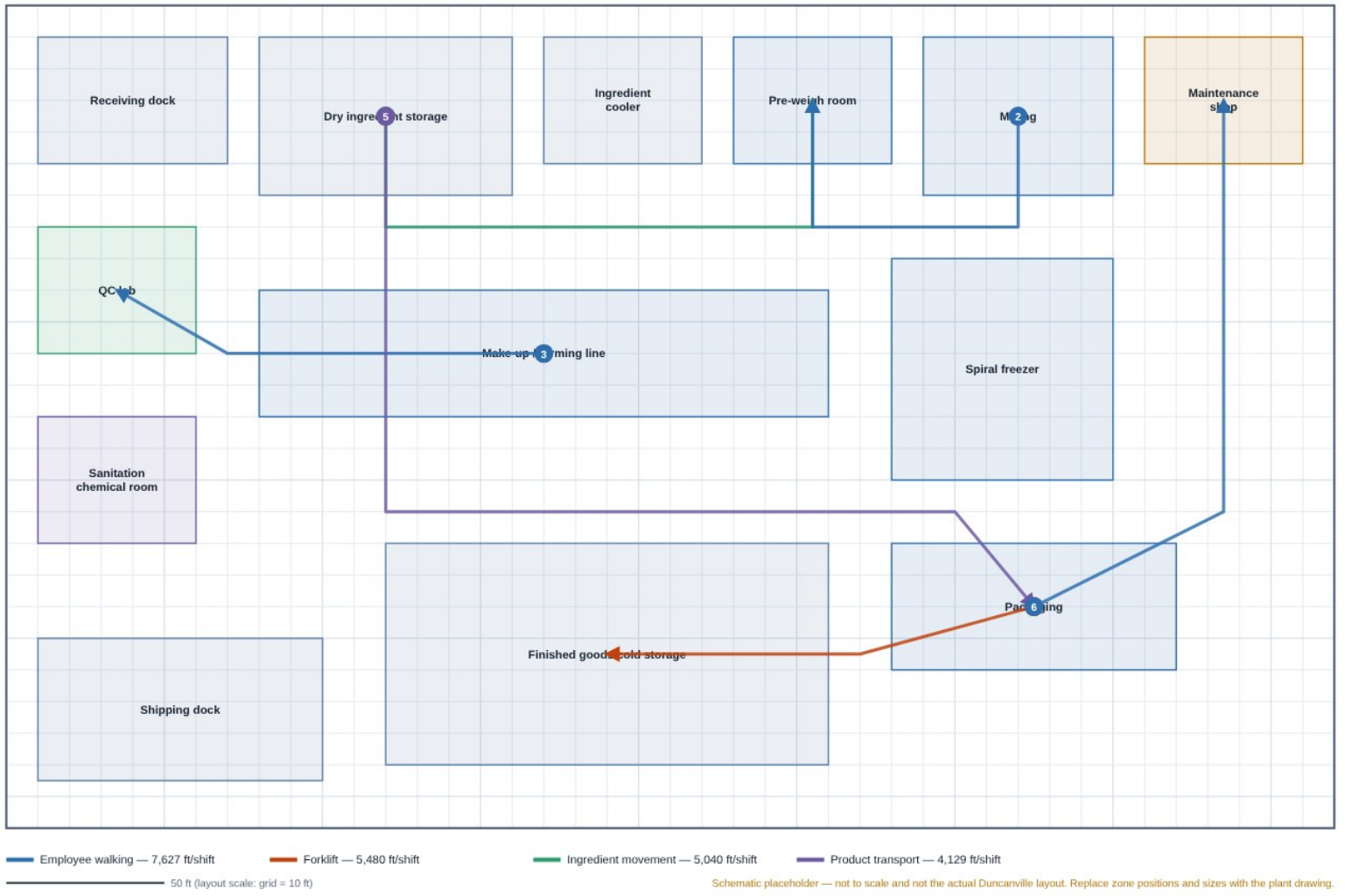


NYC = Not Yet Collected. Values marked (est.) include estimates; (incomplete) means some inputs are missing.

DEMONSTRATION DATA — NOT PLANT MEASUREMENTS

Spaghetti diagram

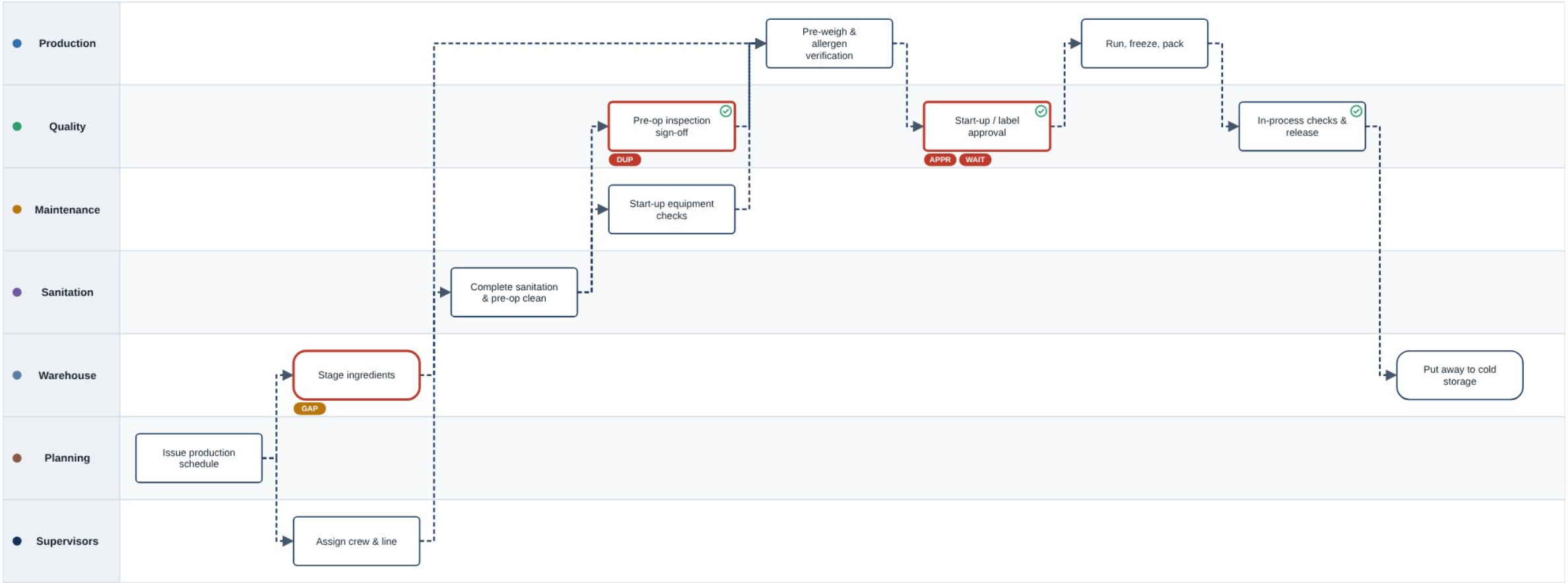
Schematic placeholder — not to scale and not the actual Duncanville layout. Replace zone positions and sizes with the plant drawing.



DEMONSTRATION DATA — NOT PLANT MEASUREMENTS

Swimlane process map

Responsibilities, handoffs and flagged approvals or delays.



Cross-department handoffs: 12 · Approvals: 3 · Flagged items: 3 · DUP = duplicated work, APPR = unnecessary approval, WAIT = delay, GAP = communication gap

DEMONSTRATION DATA — NOT PLANT MEASUREMENTS

Step data

NYC = Not Yet Collected · (E) estimate · (V) verified

Step	Applies	Cycle time (s/unit)	Processing time (value-added) (min)	Changeover time (min)	Uptime (%)	Labor (people)	Throughput (case/hr)	Scrap (%)	WIP before step (case)	OEE (%)
Raw Material Receiving	Confirmed	4 (E)	0 (E)	NYC	95 (E)	2 (E)	NYC	NYC	9,000 (E)	NYC
Ingredient Storage	Confirmed	2 (E)	0 (E)	NYC	NYC	1 (E)	NYC	NYC	14,000 (E)	NYC
Pre-Weigh	Confirmed	5.1 (E)	3 (E)	25 (E)	92 (E)	2 (E)	NYC	0.3 (E)	900 (E)	NYC
Mixing	Confirmed	5.4 (E)	14 (E)	45 (E)	88 (E)	2 (E)	NYC	0.6 (E)	400 (E)	71 (E)
Dividing / Forming	Confirmed	5.9 (E)	2 (E)	60 (E)	82 (E)	4 (E)	610 (E)	2.4 (E)	150 (E)	64 (E)
Sheeting / Lamination	Not applicable	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Proofing	Not applicable	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Baking / Par-Baking	Not applicable	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Cooling	Not applicable	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC	NYC
Spiral Freezing	Confirmed	5 (E)	32 (E)	NYC	94 (E)	1 (E)	NYC	0.2 (E)	600 (E)	NYC
Packaging	Confirmed	6.3 (E)	1 (E)	40 (E)	80 (E)	6 (E)	571 (E)	1.1 (E)	250 (E)	61 (E)
Cold Storage	Confirmed	2.5 (E)	0 (E)	NYC	NYC	1 (E)	NYC	NYC	16,000 (E)	NYC
Shipping	Confirmed	3 (E)	0 (E)	NYC	NYC	2 (E)	NYC	NYC	0 (E)	NYC

Process activity map

Total 1,478 min · NVA 1,404 min · 960 ft · 0 activities not timed

Step	Activity	Type	Time (min)	Distance (ft)	People	Class	Data
Raw Material Receiving	Wait / queue before raw material receiving	Delay	20			NVA	Estimate
Raw Material Receiving	Raw Material Receiving	Operation	8		2	NNVA	Estimate
Raw Material Receiving	Quality / food-safety check at raw material receiving	Inspection	2		1	NNVA	Estimate
Ingredient Storage	Move from previous step	Transportation	3	120	1	NVA	Estimate
Ingredient Storage	Wait / queue before ingredient storage	Delay	20			NVA	Estimate
Ingredient Storage	Held in ingredient storage	Storage	600			NVA	Estimate
Ingredient Storage	Quality / food-safety check at ingredient storage	Inspection	2		1	NNVA	Estimate
Pre-Weigh	Move from previous step	Transportation	3	120	1	NVA	Estimate
Pre-Weigh	Wait / queue before pre-weigh	Delay	20			NVA	Estimate
Pre-Weigh	Pre-Weigh	Operation	8		2	NNVA	Estimate
Pre-Weigh	Quality / food-safety check at pre-weigh	Inspection	2		1	NNVA	Estimate
Mixing	Move from previous step	Transportation	3	120	1	NVA	Estimate
Mixing	Wait / queue before mixing	Delay	20			NVA	Estimate
Mixing	Mixing	Operation	8		2	VA	Estimate
Mixing	Quality / food-safety check at mixing	Inspection	2		1	NNVA	Estimate
Dividing / Forming	Move from previous step	Transportation	3	120	1	NVA	Estimate
Dividing / Forming	Wait / queue before dividing / forming	Delay	20			NVA	Estimate
Dividing / Forming	Dividing / Forming	Operation	8		2	VA	Estimate
Dividing / Forming	Quality / food-safety check at dividing / forming	Inspection	2		1	NNVA	Estimate
Spiral Freezing	Move from previous step	Transportation	3	120	1	NVA	Estimate
Spiral Freezing	Wait / queue before spiral freezing	Delay	20			NVA	Estimate
Spiral Freezing	Spiral Freezing	Operation	8		2	VA	Estimate
Spiral Freezing	Quality / food-safety check at spiral freezing	Inspection	2		1	NNVA	Estimate
Packaging	Move from previous step	Transportation	3	120	1	NVA	Estimate
Packaging	Wait / queue before packaging	Delay	20			NVA	Estimate
Packaging	Packaging	Operation	8		2	VA	Estimate

Step	Activity	Type	Time (min)	Distance (ft)	People	Class	Data
Packaging	Quality / food-safety check at packaging	Inspection	2		1	NNVA	Estimate
Cold Storage	Move from previous step	Transportation	3	120	1	NVA	Estimate
Cold Storage	Wait / queue before cold storage	Delay	20			NVA	Estimate
Cold Storage	Held in cold storage	Storage	600			NVA	Estimate
Cold Storage	Quality / food-safety check at cold storage	Inspection	2		1	NNVA	Estimate
Shipping	Move from previous step	Transportation	3	120	1	NVA	Estimate
Shipping	Wait / queue before shipping	Delay	20			NVA	Estimate
Shipping	Shipping	Operation	8		2	NNVA	Estimate
Shipping	Quality / food-safety check at shipping	Inspection	2		1	NNVA	Estimate

Waste & opportunity register, actions, finance validation

Estimated savings are not validated until Finance confirms them.

Step	Waste	Opportunity	Est. $/yr	Benefit type	Finance status	Validated $/yr
Packaging	Waiting	Packaging is the bottleneck: frequent jams and slow changeover	$18,500	Capacity	In verification	—
Dividing / Forming	Defects	Piece-weight scrap 2.4% at forming; tighten divider set-up	$31,000	Hard savings	Method agreed with Finance	—
Mixing	Motion	Mixer operator walks to pre-weigh for every batch	$6,100	Capacity	Not started	—
Pre-Weigh	Inventory	Weighed kits staged a full shift ahead of mixing	$0	Nonfinancial	Not started	—
Packaging	Over-processing	Separate start-up label approval duplicates the pre-op check	$2,400	Capacity	Validated by Finance	$1,800

Action	Owner	Due	Status
Time 5 packaging changeovers and build SMED task list	DEMO supervisor	2026-11-23	In progress
Pareto of packaging jams from downtime log	DEMO maintenance	2026-11-10	Not started (overdue)
Agree scrap baseline and valuation method with Finance	DEMO CI	2026-11-20	Done
Trial kit cart for weighed ingredients	DEMO supervisor	2026-12-04	Not started
Combine start-up label approval into pre-op sign-off (Quality approval required)	DEMO quality	2026-11-27	Blocked

Floor-walk checklist — open items

86 items still to collect

Area	Item	How to measure	Done
Raw Material Receiving	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Raw Material Receiving	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Raw Material Receiving	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Raw Material Receiving	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Raw Material Receiving	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Raw Material Receiving	Yield (%)	Good output ÷ input (weight basis), from production records.	
Raw Material Receiving	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Raw Material Receiving	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Raw Material Receiving	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Raw Material Receiving	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Raw Material Receiving	Verify food-safety controls, allergen controls, traceability records, quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Ingredient Storage	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Ingredient Storage	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Ingredient Storage	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Ingredient Storage	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin cords.	
Ingredient Storage	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Ingredient Storage	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Ingredient Storage	Yield (%)	Good output ÷ input (weight basis), from production records.	
Ingredient Storage	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	

Area	Item	How to measure	Done
Ingredient Storage	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Ingredient Storage	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Ingredient Storage	Verify food-safety controls, allergen controls, traceability records, sanitation requirements	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Pre-Weigh	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Pre-Weigh	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Pre-Weigh	Yield (%)	Good output ÷ input (weight basis), from production records.	
Pre-Weigh	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Pre-Weigh	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Pre-Weigh	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Pre-Weigh	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Pre-Weigh	Verify allergen controls, traceability records, quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Mixing	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Mixing	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Mixing	Yield (%)	Good output ÷ input (weight basis), from production records.	
Mixing	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Mixing	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Mixing	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Mixing	Verify traceability records, quality checks, sanitation requirements	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Dividing / Forming	Yield (%)	Good output ÷ input (weight basis), from production records.	
Dividing / Forming	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Dividing / Forming	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Dividing / Forming	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Dividing / Forming	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Dividing / Forming	Verify food-safety controls, quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Spiral Freezing	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Spiral Freezing	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Spiral Freezing	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Spiral Freezing	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	

Area	Item	How to measure	Done
Spiral Freezing	Yield (%)	Good output ÷ input (weight basis), from production records.	
Spiral Freezing	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Spiral Freezing	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Spiral Freezing	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Spiral Freezing	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Spiral Freezing	Verify quality checks, sanitation requirements	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Packaging	Yield (%)	Good output ÷ input (weight basis), from production records.	
Packaging	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Packaging	Transport distance (ft/trip)	Measured route from the previous step (tape / wheel / plant drawing).	
Packaging	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Packaging	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Packaging	Verify food-safety controls, allergen controls, traceability records, quality checks	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Cold Storage	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Cold Storage	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Cold Storage	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Cold Storage	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin	cords.
Cold Storage	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	
Cold Storage	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Cold Storage	Yield (%)	Good output ÷ input (weight basis), from production records.	
Cold Storage	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Cold Storage	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Cold Storage	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Cold Storage	Verify food-safety controls, traceability records	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Shipping	Changeover time (min)	Last good unit of product A to first good unit of product B. Observe 3+ changeovers; also pull the changeover log.	
Shipping	Changeovers per week (/week)	Count from the production schedule or changeover log, last 4 weeks average.	
Shipping	Unplanned downtime (min/shift)	Downtime log, last 4 weeks, average minutes per shift; capture top 3 causes.	
Shipping	Uptime (%)	(Planned time " unplanned downtime " changeover) ÷ planned time, from lin	cords.
Shipping	Throughput (cases/hr)	Count good units over at least 1 hour of normal running, or pull from line counter.	

Area	Item	How to measure	Done
Shipping	Scrap (%)	Scrap weight ÷ input weight at the step, from the scrap log or a shift weigh-out.	
Shipping	Yield (%)	Good output ÷ input (weight basis), from production records.	
Shipping	Operator walking (ft/shift)	From the spaghetti diagram: measured route length × trips per shift (or pedometer).	
Shipping	Waiting time before step (min)	Observed hold / queue time. Used for lead time only where WIP was not counted.	
Shipping	OEE (%)	Availability × Performance × Quality from line data. Equipment steps only.	
Shipping	Operating cost ($/hr)	From Finance: labor + utilities + overhead rate for the step. Do not estimate without Finance.	
Shipping	Verify food-safety controls, traceability records	Compare the template notes against the food safety plan, allergen matrix and QC records; correct the step notes.	
Spaghetti diagram	Replace placeholder layout with plant drawing dimensions	Use the facility drawing; measure zone positions in feet.	
Spaghetti diagram	Trace walking, forklift, ingredient and product routes for one full shift	Follow each role with a layout printout; count trips; measure routes with a measuring wheel.	
Swimlane	Confirm the handoff and approval sequence with each department	Short interviews with Planning, Warehouse, Sanitation, Quality, Maintenance and the supervisors; mark duplicated work and waits.	
Food safety	Observe one allergen changeover and the sanitation verification	Record steps, times and sign-offs; compare with the allergen changeover procedure.	

Formulas

How each result is calculated

Calculation	Formula	Unit
Available time per day	Shifts/day × (Shift length " Planned breaks)	min/day
Takt time	Available time per day × 60 ÷ Customer demand per day	s/unit
Effective cycle time (per step)	Cycle time ÷ Uptime. Where uptime is not collected, cycle time is used and the result is marked Estimate.	s/unit
Bottleneck	Step with the highest effective cycle time among steps that apply to the line	
Line capacity	3600 ÷ Bottleneck effective cycle time	units/hr
Queue time (per step)	WIP before step ÷ Demand per day × Available time per day. If WIP was not counted, the observed Waiting time is used.	min
Total lead time	£ Queue time + £ Processing time over applicable steps	min
Value-added time	£ Processing time of steps classified Value-added	min
Non-value-added time	Total lead time " Value-added time	min
Process cycle efficiency (PCE)	Value-added time ÷ Total lead time	%
Rolled throughput yield	(Yield) over steps; where yield is missing, (1 " Scrap) is used	%
Changeover time per week	£ (Changeover time × Changeovers per week) ÷ 60	hr/week
Route distance	£ segment lengths of the drawn route × layout scale (1 unit = 1 ft)	ft/trip
Travel time per shift	Route distance × Trips per shift ÷ Speed for the movement category	min/shift
Annual travel labor hours	Travel time per shift × People × Shifts/day × Operating days/year ÷ 60	hr/yr
Annual travel labor cost	Annual travel labor hours × Loaded labor rate	$/yr
Estimated annual labor savings (travel)	Current annual travel labor cost " Future annual travel labor cost	$/yr
Changeover reduction	(Current " Future changeover hours/week) × Operating weeks/year; labor value = £ hours × step lab	hr/yr te
Scrap reduction opportunity	(Future RTY " Current RTY) × Demand/day × Operating days/year × Material cost/unit (valued at cos	$/yr
Throughput improvement potential	(Future capacity " Current capacity) ÷ Current capacity	%
Labor productivity	Line capacity ÷ Total labor on the line	units/labor-hr
PAM totals	£ activity time, distance and people-minutes by activity type (O, T, I, D, S); NVA time = £ time of a	min / ft ked N